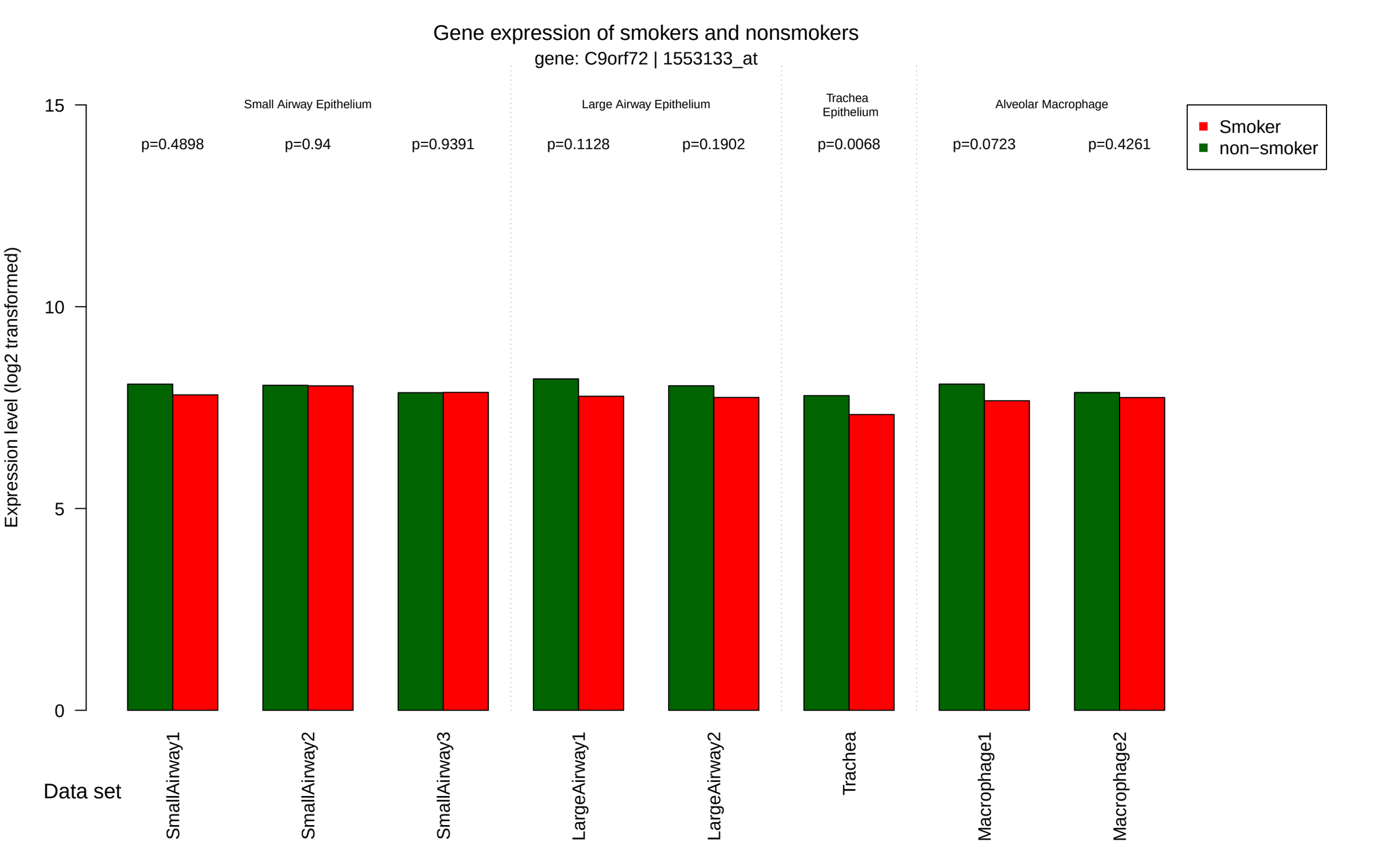
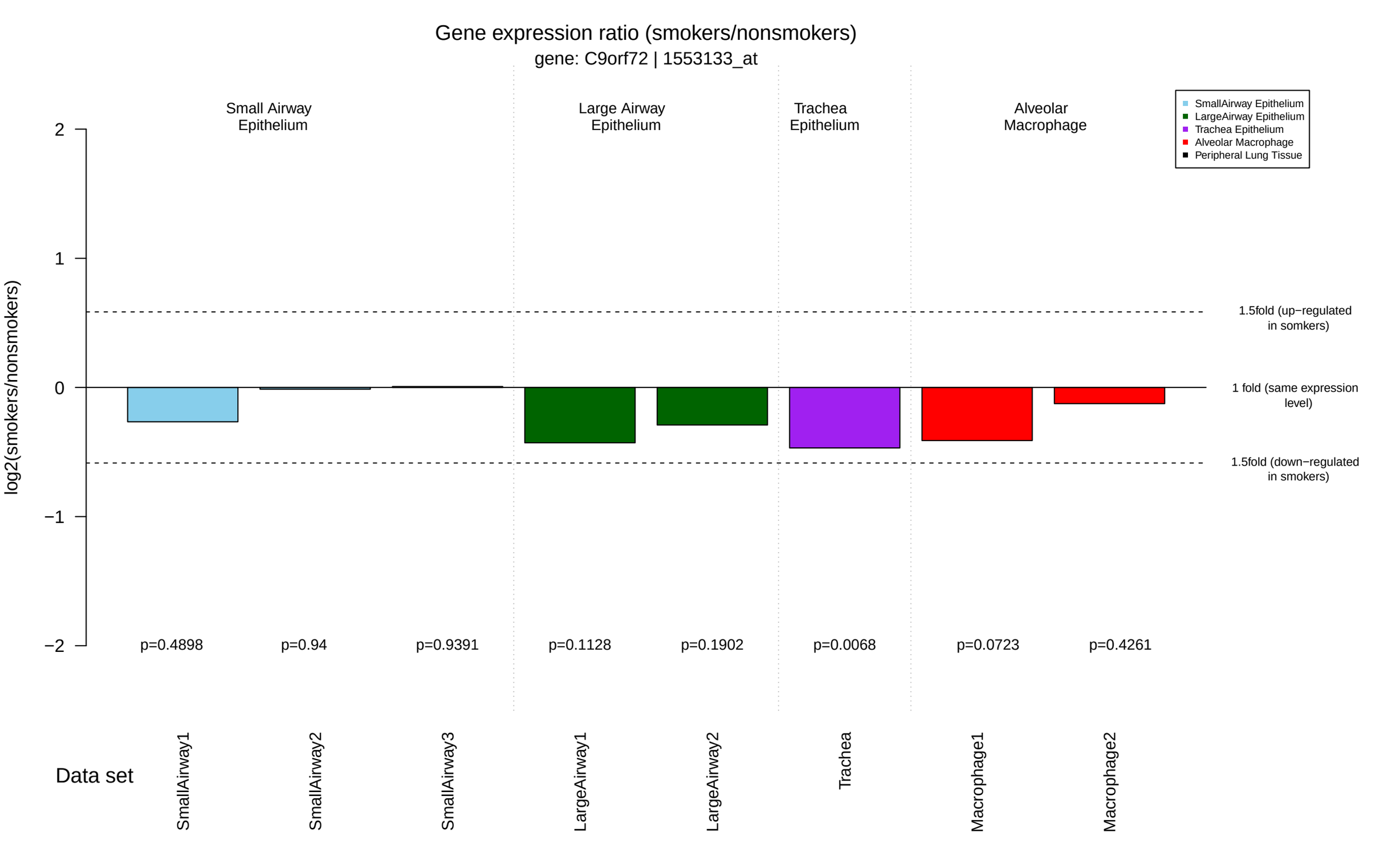
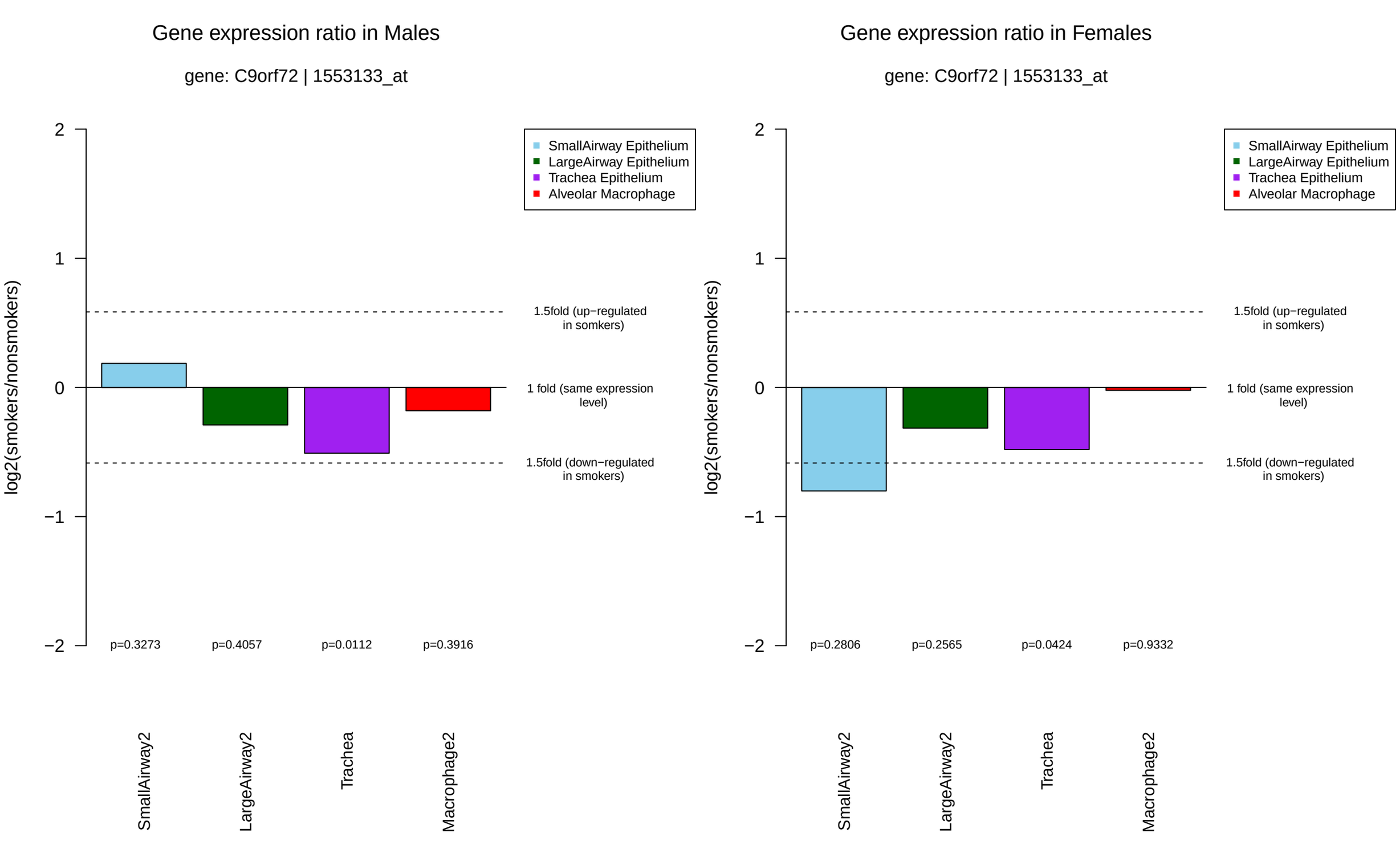
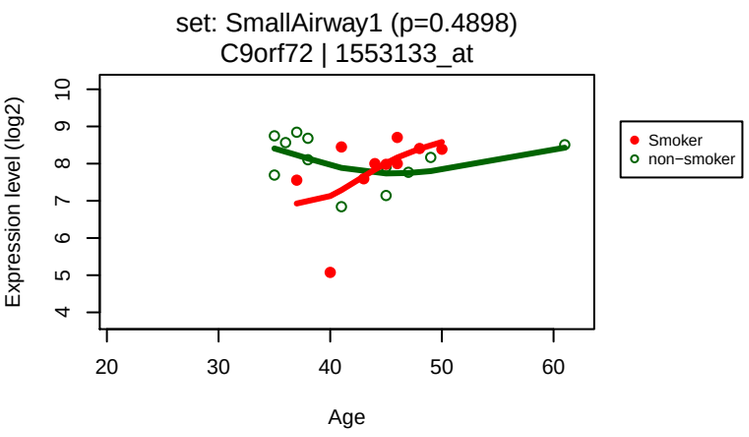
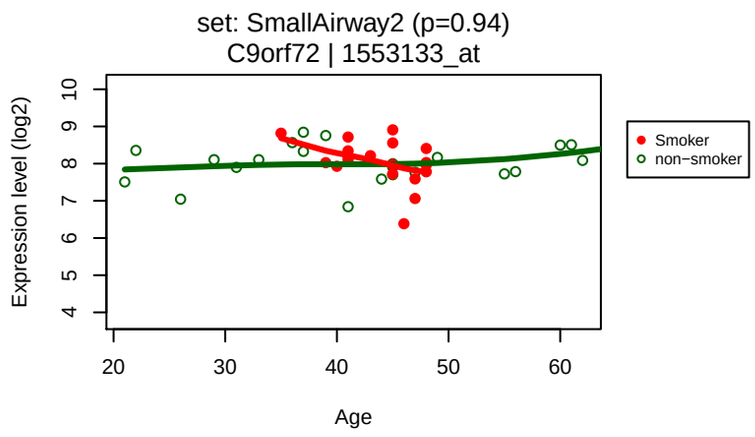
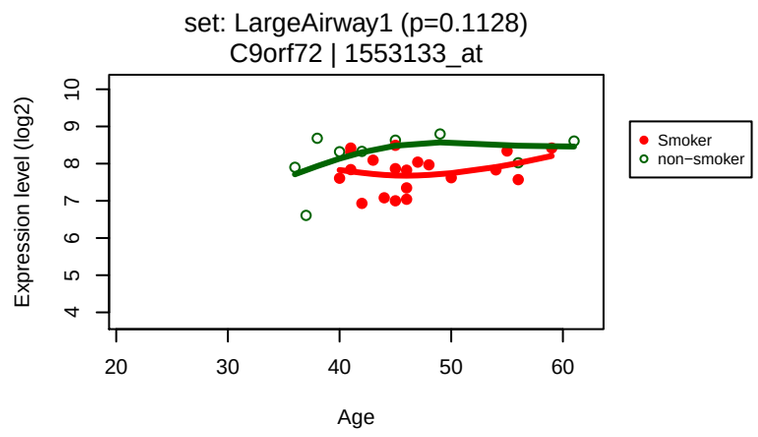

Gene expression of smokers and nonsmokers
gene: C9orf72 | 1553133_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4898
p=0.94
p=0.9391
p=0.1128
p=0.1902
p=0.0068
p=0.0723
p=0.4261
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

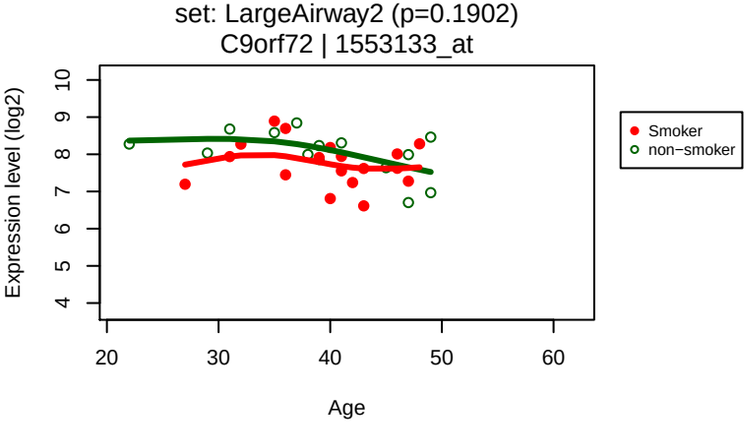

Gene expression ratio (smokers/nonsmokers)
gene: C9orf72 | 1553133_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4898
p=0.94
p=0.9391
p=0.1128
p=0.1902
p=0.0068
p=0.0723
p=0.4261
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

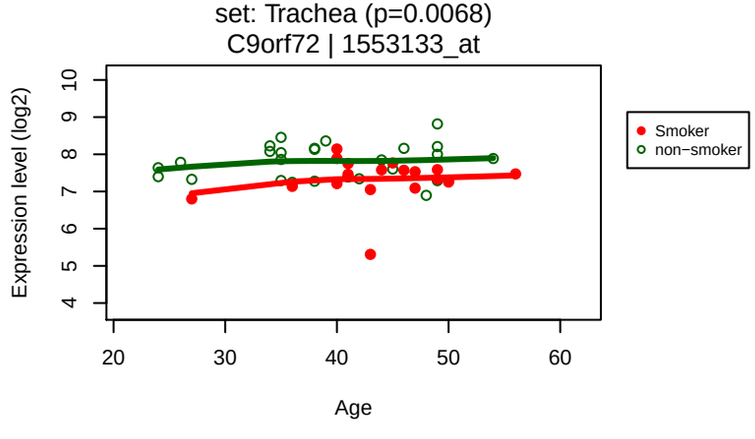

Gene expression ratio in Males
gene: C9orf72 | 1553133_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3273
p=0.4057
p=0.0112
p=0.3916
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: C9orf72 | 1553133_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2806
p=0.2565
p=0.0424
p=0.9332
SmallAirway2
LargeAirway2
Trachea
Macrophage2

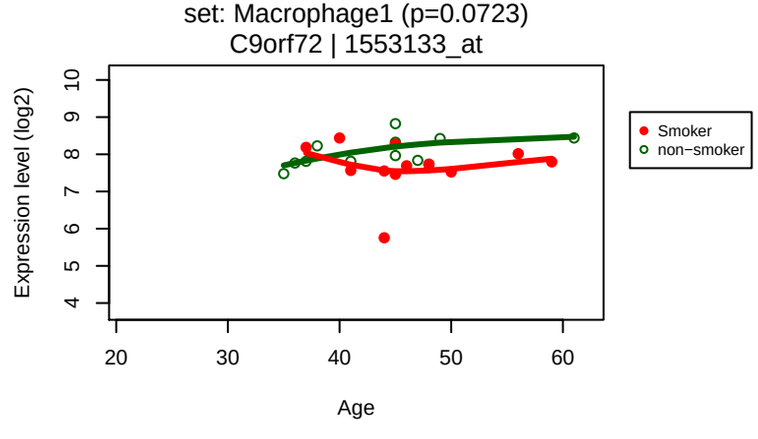

set: SmallAirway1 (p=0.4898)
C9orf72 | 1553133_at
Expression level (log2)
Age
Smoker
non−smoker

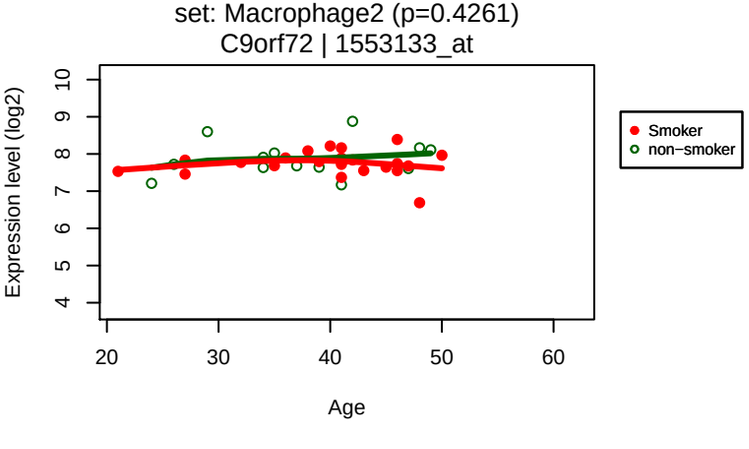

set: SmallAirway2 (p=0.94)
C9orf72 | 1553133_at
Expression level (log2)
Age
Smoker
non−smoker

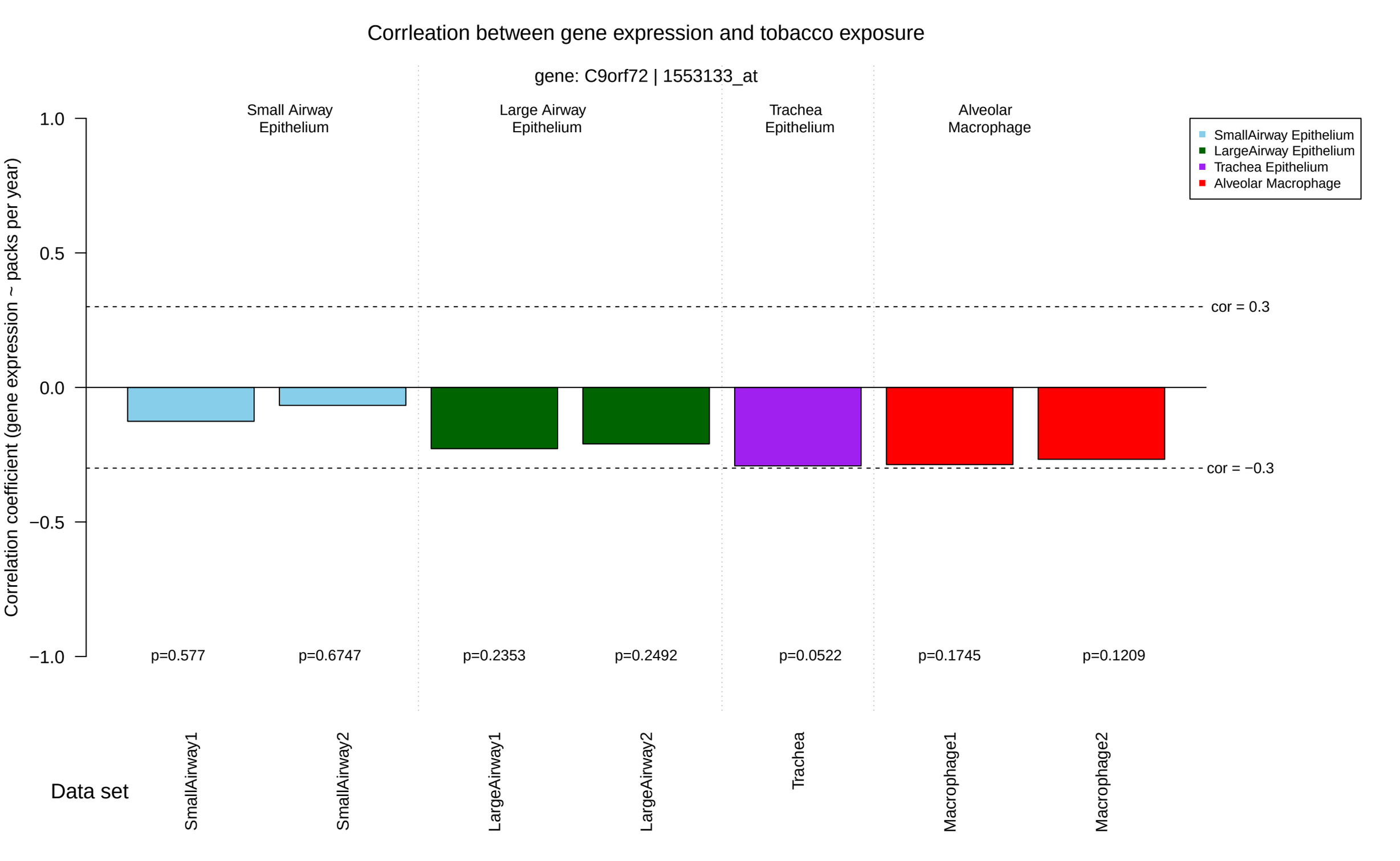

set: LargeAirway1 (p=0.1128)
C9orf72 | 1553133_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1902)
C9orf72 | 1553133_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0068)
C9orf72 | 1553133_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0723)
C9orf72 | 1553133_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4261)
C9orf72 | 1553133_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: C9orf72 | 1553133_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.577
p=0.6747
p=0.2353
p=0.2492
p=0.0522
p=0.1745
p=0.1209
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2